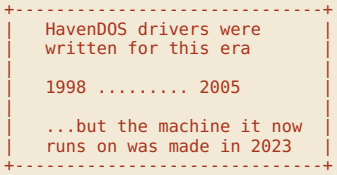
TECHHAVEN STUDIOS

ROAD TO 1.0

HavenDOS — what stands between v0.8.1 and a 1.0 release



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BASIS v0.8.1, 787 checks across 16 suites
STATUS PROPOSAL — not yet agreed

1. The one-line version

Three separate complaints — installing fails on real hardware, modern features are missing, BOOT Chat does not work — are one problem:

> THE ROOT CAUSE
>
> Every driver in HavenDOS targets hardware from roughly 1998 to 2005. The machine it is now being tested on was made in 2023. Nothing is broken; the drivers are simply for a different era of computer.

That is good news, because it means there is one coherent goal rather than a scattered bug list. 1.0 is the release where HavenDOS works on a computer made in the last decade.

2. What is actually supported today

Read from the v0.8.1 tree, not from memory:

Area	HavenDOS drives	A 2023 laptop has
Disk	Legacy IDE, VirtIO, NVMe READ ONLY	NVMe, or SATA in AHCI mode
USB	UHCI (USB 1.1, 1996)	xHCI (USB 3)
Network	RTL8139, 7 e1000 IDs	RTL8168, or Wi-Fi only
Audio	— (AC'97 planned)	Intel HD Audio
Filesystem	FAT12, FAT16	FAT32 ESP, NTFS
Partitions	MBR write, GPT READ ONLY	GPT
Boot	BIOS GRUB to MBR	UEFI only, no CSM

Every row in the middle column is from before 2006. Every row in the right column is what the test machine actually has.

3. Why installing fails

It is not one bug. The installer is architecturally BIOS-era, and a UEFI machine rejects it at four independent points:

1. It cannot write to the disk. NVMe is read-only by design, and there is no AHCI driver at all, so a SATA laptop is invisible too.
2. It writes an MBR. Step 1 zeroes LBA 0-63 and the last 33 sectors, then writes a fresh MBR partition table.
3. It installs BIOS GRUB. A UEFI-only machine will not execute it, so even a successful install would not boot.
4. It cannot reach the ESP. An EFI System Partition is FAT32, and HavenDOS has FAT12 and FAT16 only.

> NOT PREVIOUSLY ON ANY LIST
> Point 4 was found while writing this document. FAT32 is a hard prerequisite for any UEFI install and no plan so far has included it.

4. Why BOOT Chat does not work

Also not a chat bug. The protocol, the server and the client are fine. There is no network interface.

```
havendos> ifconfig
  no NIC detected

HavenDOS matches exactly two network chips:
  10EC:8139   RTL8139     (1998, 10/100)
  8086:xxxx   e1000       (7 IDs, 2002-2004)
```

FIGURE 4-1. The whole of HavenDOS network hardware support.

A 2023 HP 15 has a Realtek RTL8168 gigabit part, which shares a vendor ID with the RTL8139 and nothing else, or it has no Ethernet socket at all and only Wi-Fi. Either way the chat client has nothing to open a socket on.

5. What 1.0 should mean

A proposed definition, to be argued with:

> PROPOSED
> HavenDOS 1.0 installs itself onto a computer made in the last ten years, boots from it, reaches the network, and does so without damaging whatever was on the machine beforehand.

Everything below is scoped to that sentence. Anything that does not serve it is explicitly out, and section 9 says so.

6. The plan

Ordered by what unlocks the most, not by what is most interesting. Each phase is useful on its own — none of them is a prerequisite that pays off only later.

Phase 1 — xHCI BIGGEST UNLOCK

USB 3 host controller. This is first because it unlocks three separate things that are each wanted for their own reasons:

- USB mass storage — the dual-boot route you chose, and the music player's next step. One driver, two goals.
- USB keyboard and mouse — a desktop with no PS/2 controller is currently unusable. The HP's internal keyboard works only because its firmware emulates an 8042.
- Everything else USB, later.

The layer above the controller already exists: Bulk-Only Transport and SCSI sit on `uhci_control()` and `uhci_bulk()`, and an xHCI driver replaces only those two. Large — xHCI is a considerably more complex controller than UHCI: rings, contexts, a command queue.

Phase 2 — AHCI

SATA in native mode, which is every laptop and desktop from roughly 2008 to 2020 and many still sold. Wider machine coverage per line of code than NVMe, and it also fixes update, which cannot read the CD on a Q35 machine today. Medium.

Phase 3 — FAT32

Read and write. Needed to touch an ESP at all, so no UEFI install is possible without it. Also the format of most USB sticks, so Phase 1 wants it too. Medium — the cluster logic is close to FAT16; the directory and FSInfo handling is not.

Phase 4 — The additive installer

The piece that makes dual boot real, and the one with the highest consequences if it is wrong.

- GPT write, additive only. Add one entry to free space, recompute both CRCs, write both copies. Never rewrite the table, never zero a header.
- ESP install. Mount the existing FAT32 ESP and add a HavenDOS loader beside what is already there.
- Boot registration. Simplest path is the removable-media fallback; a real NVRAM boot entry can come later.
- Self-hosting install, which is the two-pass problem already recorded in the handoff.

> **CAUTION**
> This phase writes to a partition table on a machine with another operating system on it. The current installer destroys a GPT by design; the new one must be provably additive. Nothing here ships without a test that checksums a Windows-shaped disk before and after.

Phase 5 — RTL8168

The common Realtek gigabit part. Restores BOOT Chat and everything else networked on any machine with an Ethernet socket. Medium.

Phase 6 — Intel HD Audio

The music player was planned around AC'97, which is as dead as the rest of the list. On the test machine the audio controller is Intel HDA. The WAV and MP3 work above it is unchanged. Medium-large.

Phase 7 — Finish the console

Smaller, already-known items that make the system feel finished:

- Desktop apps lay out for 80×25 and draw in the corner of anything larger. The console now guarantees 80×25 centred, which is the cheap half; the apps are the rest.
- The mouse is clamped to column 79 and row 24, so above 80×25 the pointer cannot reach the start button.
- Dotfiles on FAT16 become blank-basename directory entries.
- Some themes draw the menu separators invisibly.

7. Dependencies

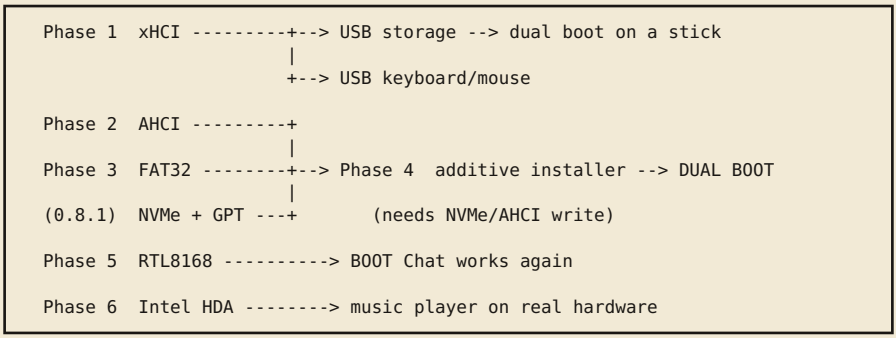

FIGURE 7-1. What blocks what. Phases 1, 2, 3 and 5 are independent of each other and can be done in any order.

8. Already done, v0.8.1

- NVMe read, GPT read, and the installer refusing both at three layers.
- Framebuffers above 4 GB paged in with PSE-36, which is what made the machine boot at all.
- Live media: the ISO probes no disk and mounts nothing.
- The console honours its requested grid and centres it on any panel.
- One app registry; the start menu grew from 23 entries to 45.
- The command reference is build-checked against the dispatcher.

9. Deliberately not in 1.0

Left out	Why
Wi-Fi	Needs signed firmware blobs per chipset and a full 802.11 MAC. Out of reach for a solo bare-metal kernel, and the honest answer is Ethernet or nothing.
NTFS	Reading a Windows partition is not needed to live beside one.
Secure Boot	Requires a signed loader. Users can turn it off; a 1.0 need not solve it.
Multitasking	Orthogonal to running on modern hardware, and large. ps already says so.
Writing to NVMe/ AHCI before Phase 4	A write path with no additive installer behind it is a loaded gun in the tree.

10. What I need from you

Two things would sharpen this a lot, and I did not want to guess:

1. Which machines did you try to install on, and what happened on each? “A lot of real hardware” could be four UEFI laptops, in which case Phase 4 is everything — or a mix including older BIOS desktops, in which case AHCI moves up and something else is wrong too.
2. Which modern features do you mean? The list above is my inference from the source. If you meant something specific that is not on it, it is missing from the plan.

Neither blocks starting: Phase 1 is right regardless of the answers.

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Proposal only. No phase has been agreed or started.